In constant currency, total margin and total margin as a percentage of revenues for our hardware segment increased in fiscal 2019 due to lower expenses .

Services Business

We offer services to customers and partners to help to maximize the performance of their investments in Oracle applications and infrastructure technologies. Services revenues are generally recognized as the services are performed. The cost of providing our services consists primarily of personnel related expenses, technology infrastructure expenditures, facilities expenses and external contractor expenses.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Services Revenues:				
Americas	$ 1,576	-5%	-3%	$ 1,654
EMEA	1,021	-2%	2%	1,046
Asia Pacific	643	-7%	-4%	695
Total revenues	3,240	-5%	-2%	3,395
Total Expenses (1)	2,703	-1%	2%	2,729
Total Margin	$ 537	-19%	-18%	$ 666
Total Margin %	17%			20%
% Revenues by Geography:				
Americas	49%			49%
EMEA	31%			31%
Asia Pacific	20%			20%

(1) Excludes stock-based compensation and certain expense allocations. Also excludes amortization of intangible assets and certain other GAAP-based expenses, which were not allocated to our operating segment results for purposes of reporting to and review by our CODMs, as further described under "Presentation of Operating Segments and Other Financial Information" above.

Excluding the effects of currency rate fluctuations, our total services revenues decreased in fiscal 2019 relative to fiscal 2018 primarily due to revenue declines in our education services and, to a lesser extent, our consulting services. During fiscal 2019, constant currency increases in our EMEA-based services revenues were offset by constant currency services revenue decreases in the Americas and the Asia Pacific regions.

In constant currency, total services expenses increased in fiscal 2019 compared to fiscal 2018 primarily due to an increase in employee related expenses and external contractor expenses associated with investments in our consulting services that support our cloud offerings. In constant currency, total margin and total margin as a percentage of total services revenues decreased during fiscal 2019 relative to fiscal 2018 due to decreased revenues and increased expenses for this business.

Research and Development Expenses: Research and development expenses consist primarily of personnel related expenditures. We intend to continue to invest significantly in our research and development efforts because, in our judgment, they are essential to maintaining our competitive position.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Research and development (1)	$ 5,063	-2%	0%	$ 5,163
Stock-based compensation	963	5%	5%	921
Total expenses	$ 6,026	-1%	0%	$ 6,084
% of Total Revenues	15%			15%

(1) Excluding stock-based compensation